

AC-M: Metallorganik - Zusammenfassung

Carbonyle & Cp-Verbindungen

Klassische Koordinationsverbindungen:

- Werner-Komplexe; O-, N-, (P-, S-) - Liganden, hart, gute σ -Donoren
- kleine LF-Aufspaltung, oft paramagnetisch (high spin)
- hoher elektrost. Bindungsanteil
- typische OZ von M: +II, +III

Metallorganische Komplexe:

- C-Liganden (weich), σ -Donor / π -Akzeptorcharakter
- hohe LF-Aufspaltung, diamagnetisch (low spin)
- überwiegend kovalent $\rightarrow$ 18 VE-Regel
- niedrige OZ ≤ 2 , insbesondere 0

Binäre Carbonyle: $V(CO)_6$, $Cr(CO)_6$, $Mn_2(CO)_{10}$, $Fe_2(CO)_9$, $Co_2(CO)_8$, $Ni_2(CO)_2$
 IR: $M^n \rightarrow n < 0: \nu(CO)$ niedrig, $n = 0: \nu(CO)$ mittel, $n > 0: \nu(CO)$ hoch

CO: kein Dipolmoment $\rightarrow$ schlechterer σ -Donor als NH_3

Carbonylkomplexe: CO-Ligand leicht nukleophil angreifbar $\rightarrow$ Aktivierung durch NaOH

Aquakomplexe: 18 VE nur bei Eisen relevant (Werner-Komplexe)

$NaCp + H_2O \rightarrow NaOH + HCp \rightarrow$ aber Ferrocen wasserstabil $\Rightarrow$ kov. Bindung

Ferrocen: el. Substitution schneller als bei Benzol $\rightarrow \pi$ -e-Dichte höher / C-Atom (nicht bei zu starken Ox.-Bed.)

Aminoferrocen stärker basisch als Anilin $\rightarrow$ e⁻-reiches N-Atom

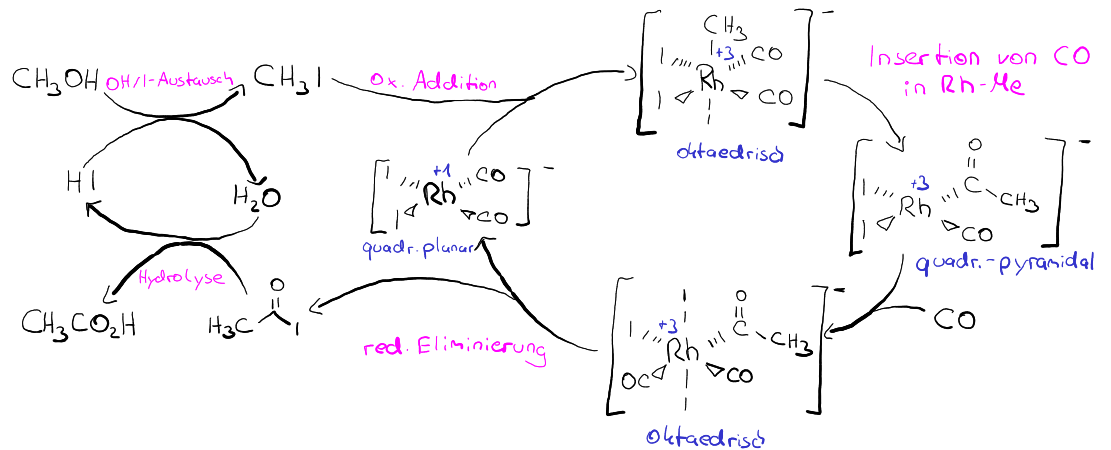
Cobaltocen: paramagnetisch $\rightarrow d^7$

keine el. Subst. möglich $\rightarrow$ immer Redox

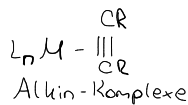
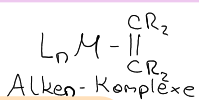
H_2O : π -Donor, hs; CO: π -Akzeptor, ls; η^5 -Cp-Liganden stabilisieren pos. OZ, CO neg. OZ

$\hookrightarrow$ HSAB?

Monsanto-Verfahren



Olefin-Komplexe

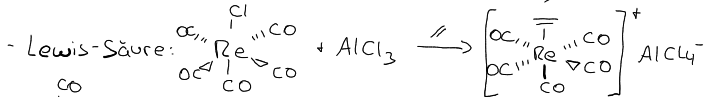
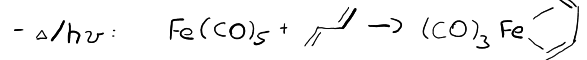
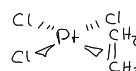
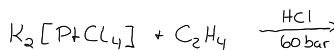


η^2 -Bindung über π -System

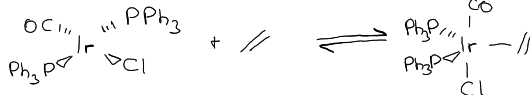
Ligandenfeldaufspaltung Olefine < Carbonyle

Synthesen

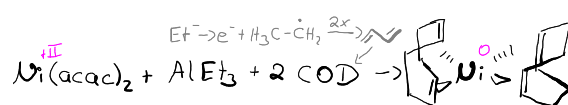
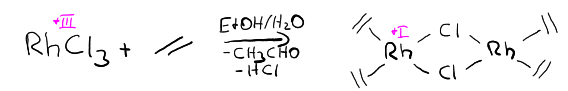
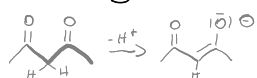
Substitutionen



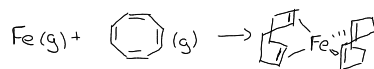
Addition an koordin. unges. Komplexe



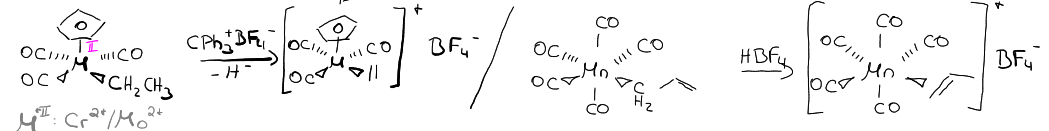
Olefinierung + Red. Metall



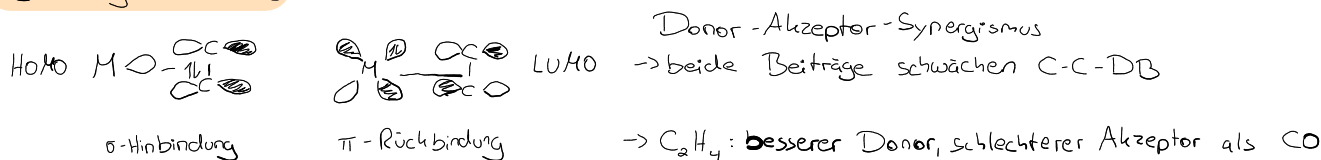
Metallatom-Ligand-Cokondensation



Erzeugung Olefinligand durch Umwandlung an M

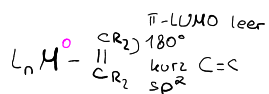


Bindungsverhältnisse

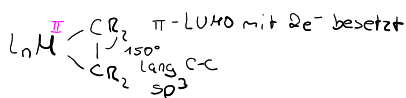


Grenzfälle:

Metall-Olefin-Addukt



Metallacyclopropan $\rightarrow$ vollständige Rückbindung Stärkung π -Akzeptoranteil



π -basische $L_n M$: niedrige OZ an M, anionische Ladung

gute Co-Liganden (CP^+ , PR_3 , Hal^-)

e^- -ziehende Substituenten an Olefin (F , CF_3 , R , NO_2 , CN)

reine Donorbindung

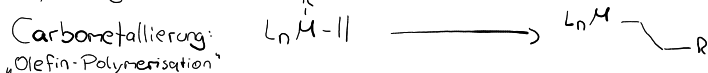
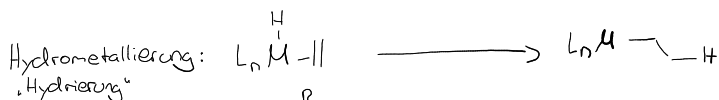
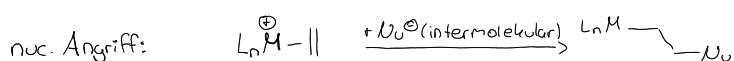
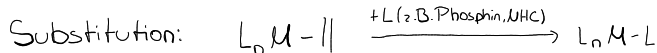
maximale Akzeptorbindung

$\rightarrow$ NMR: $\delta(1H)$ Hochfeldverschiebung je stärker Rückbindung Olefin-M, J_{CH} abhängig von s-Charakter $\rightarrow$ ^{13}C -Satelliten

Koaleszenz: geringere Feldstärke, um bei gleichem T Koaleszenz zu erreichen

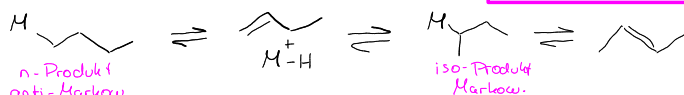
^{13}C $sp^3 \sim 125$ Hz
 $sp^2 \sim 160$ Hz
 $sp \sim 250$ Hz

Reaktivität



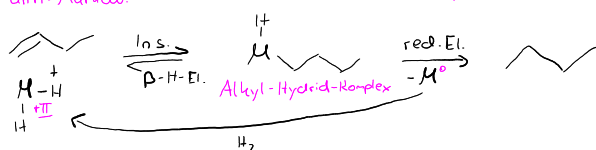
Insertion:

• Monohydrid-Komplexe
 rev. Insertion, β -H-Eliminierung



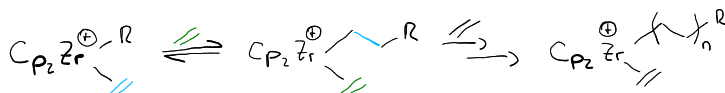
M-H kann Olefinisomerisierung katalysieren

• Dihydrid-Komplexe



Olefin-Hydrierung

in M-C:



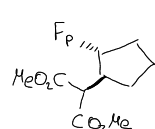
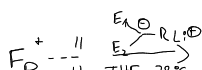
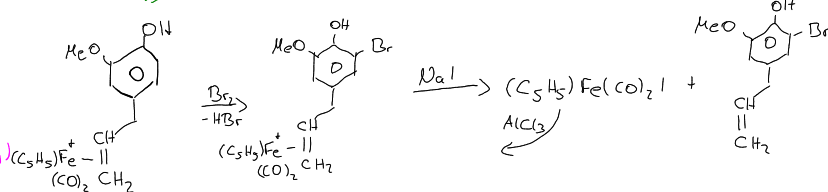
Olefin-Polymerisation

Additionsreaktion:

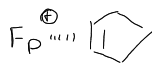
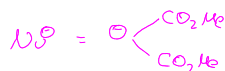
Fries Olefin: A_E

Koord. an M aktiviert Olefin für A_N (Umpolung)

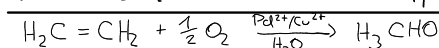
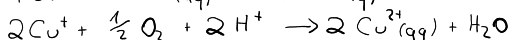
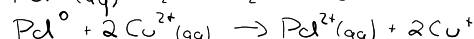
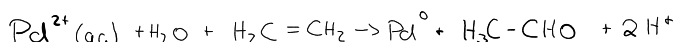
$\rightarrow L_n M$ als Schutzgruppe



$R = H$ $E_1 = CO_2Me$ $E_2 = CO_2Me$ 98% $OC(OMe) \equiv Fp^+ - II$
 H CO_2Et $C(OMe)$ 80%



Wacker-Prozess



β -H-/red. Eliminierung

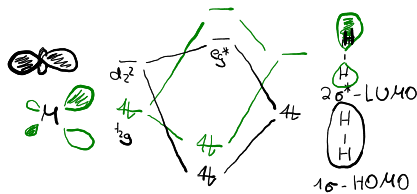
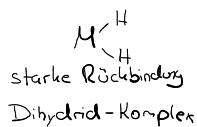
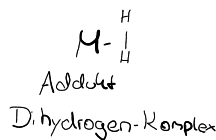
Insertion

β -H-Eliminierung

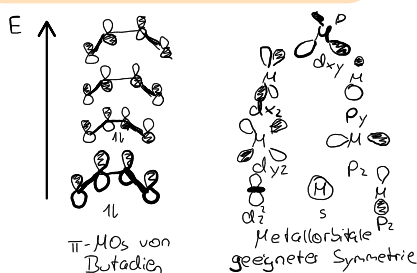
Substitution

nucleophiler Angriff

H₂-Komplexe



Komplexe konjugierter Diene

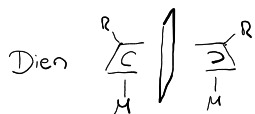
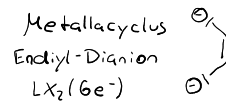
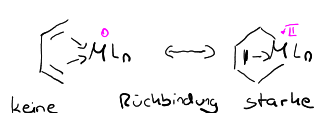
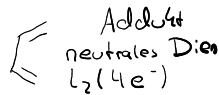


WW HOMO/LUMO Dien mit M

C-C (intern) → kürzer

C-C (terminal) → länger

σ-Bindung (Rückbindung)
π-Bindung (Rückbindung)
π-Bindung (Hinbindung)
σ-Bindung (Hinbindung)

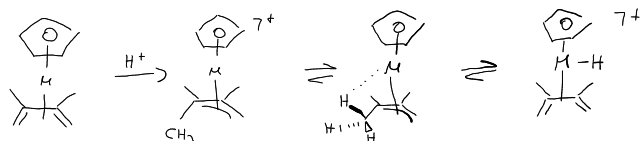
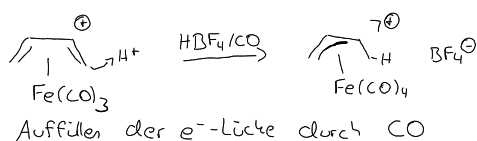


Konfigurationsstabil für R ≠ H
Enantiomere isolierbar
späte ÜM



Konfigurationsstabil für R = H
Enantiomere nicht isolierbar

Reaktionen: A_F von H⁺

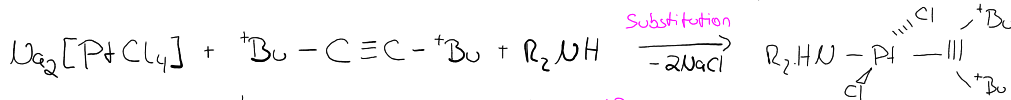


Auffüllen der e⁻-Lücke durch agostische WW/ox. Add.

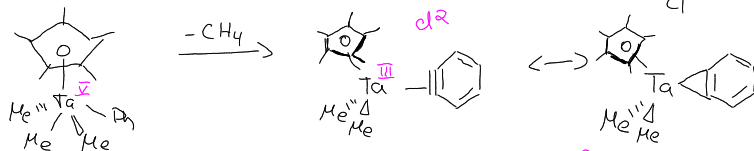
Alkin-Komplexe



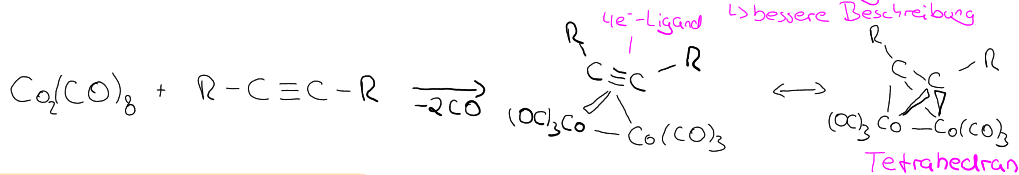
- Bdg ähnlich Olefine, 4e⁻ in π-System
- größere strukturelle Vielfalt
- stabilere Komplexe als mit Olefinen



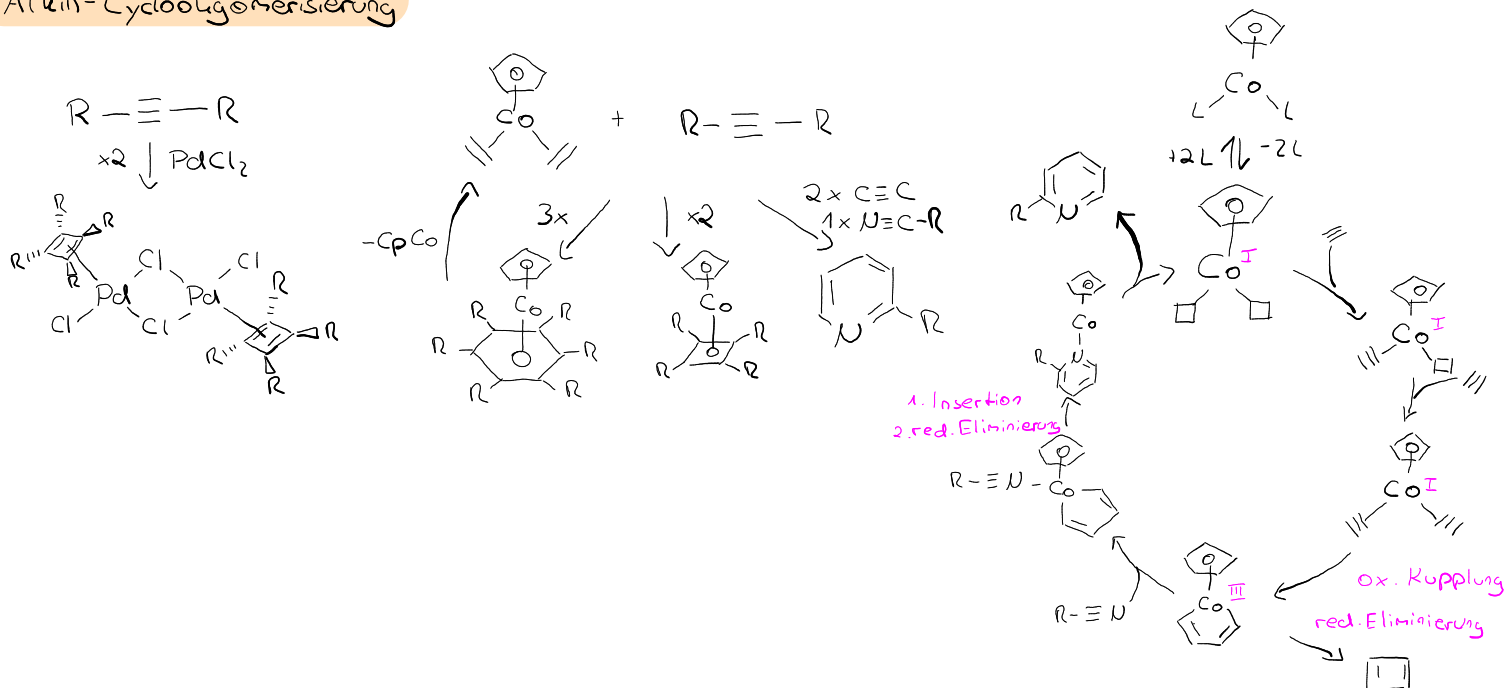
Schrock



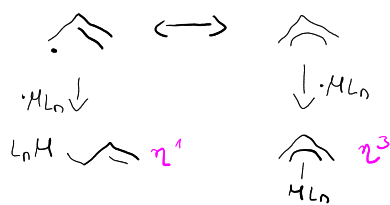
Stabilisierung Anin
↳ bessere Beschreibung



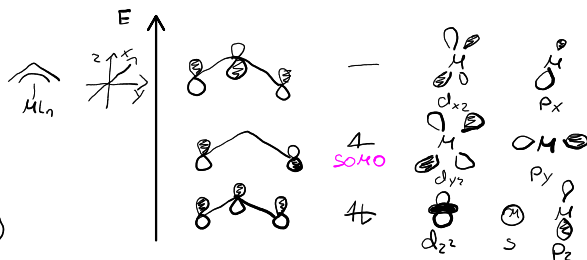
Alkin-Cyclooligomerisierung



Bindungsverhältnisse

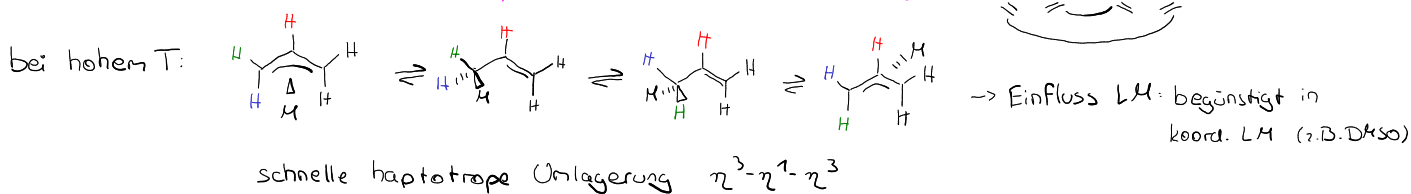
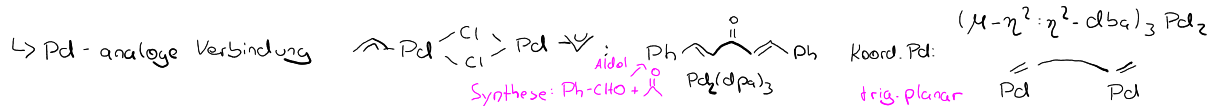
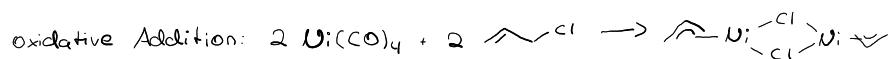
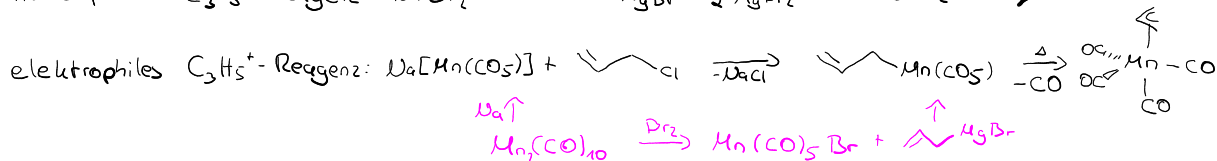
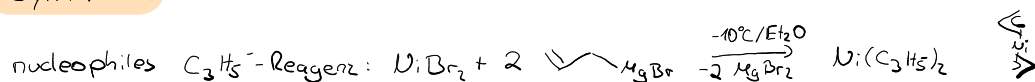


- HG-Metalle
- O₂ & H₂O empfindlich
- vorw. ionisch
- Reaktion als C₃H₅⁻-Anion
- nucleophil
- NG-Metalle (bes. späte)
- überwiegend kovalent
- unempfindlich gegen O₂ & H₂O
- Reaktion als C₃H₅⁺-Kation

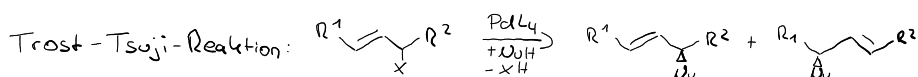


- elektropositives M : Reduktion unter Bildung $C_3H_5^-$
- UM : Bildung kov. Bindung

Synthesen



Reaktionen

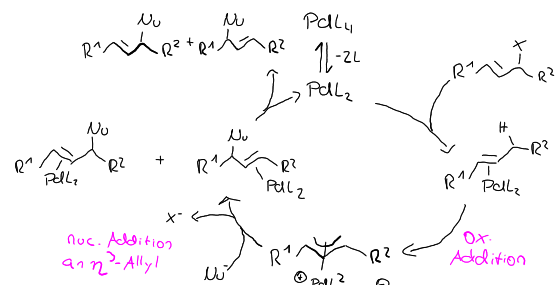


$X: OAc, OCO, R, OTs$

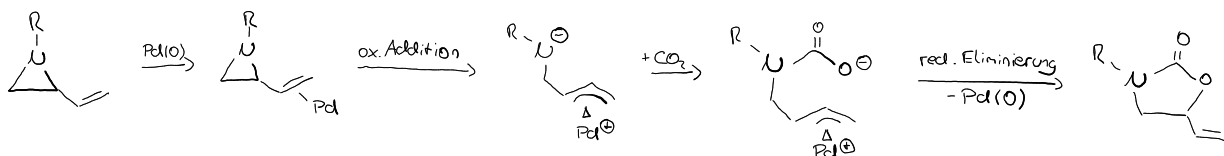
NU: Carbanionen, Amine, Phenole

Kontrolle Stereo-/Regioselektivität
durch geeignete Liganden

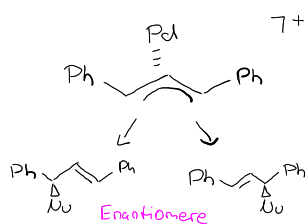
$R^1 = R^2 = H$ identisch
 $R^1 = R^2 \neq H$ enantiomer
 $R^1 \neq R^2$ diastereomer



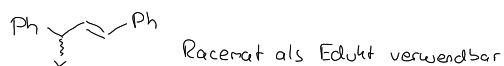
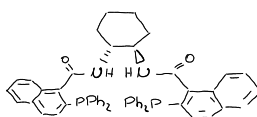
Anwendung:



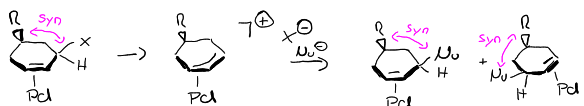
Reaktivität



chiraler P,P-Ligand ermöglicht Diskriminierung der enantiotopen Positionen $\rightarrow$ nun diastereotop

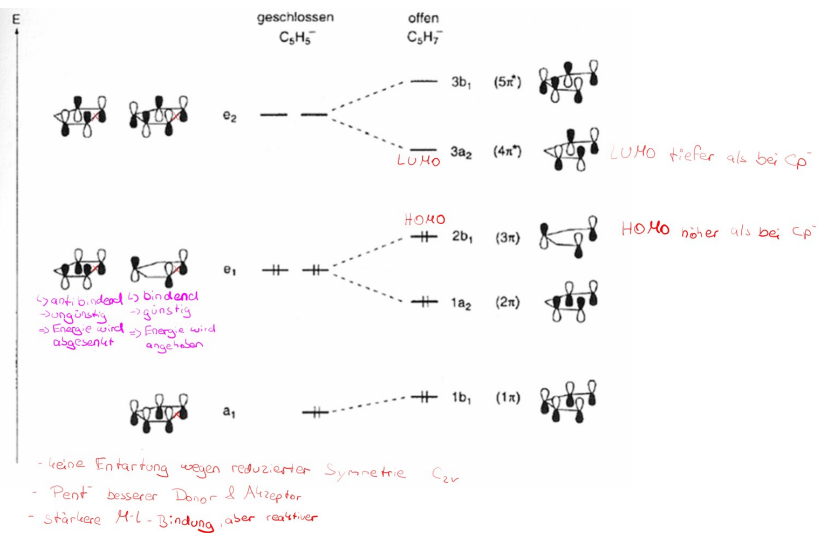
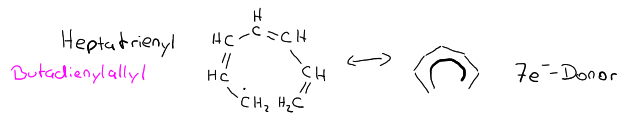
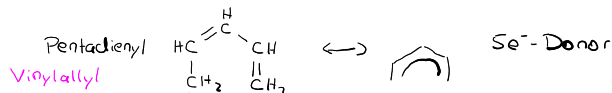
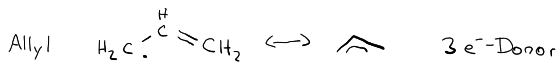


Nachweis stereochemischer Verlauf:

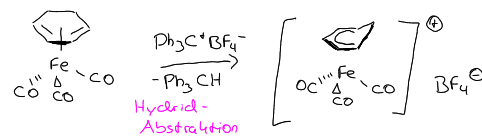
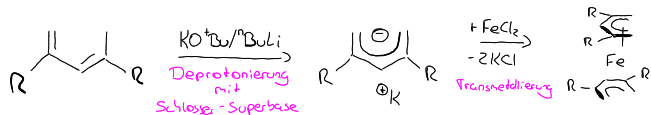


Doppelte Inversion = Retention

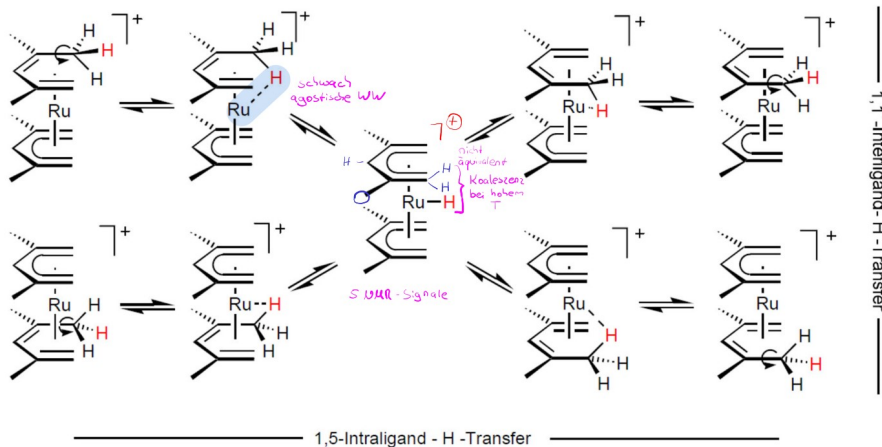
Vinyllog Allylsysteme



Pentadienylkomplexe



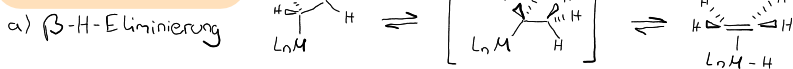
Dynamisches Verhalten in prot., offenen Ruthenocen



- C-H-Bindung als 2e-Ligand für M: agostische Wechselwirkung (Vgl. H_2 -Komplexe)
- mit D^+ H/D-Austausch an den terminalen CH_2 -Gruppen

Komplexe mit μ -C-Einfachbindungen (Alkyl, Aryl)

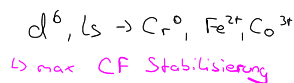
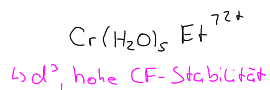
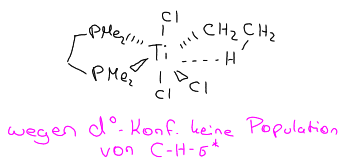
Zerfallsreaktionen



Blockierung:

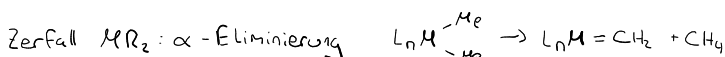
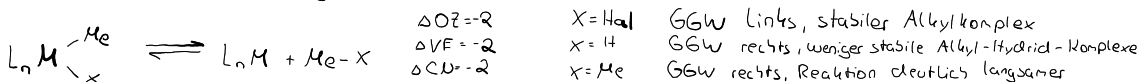
- Alkylreste ohne β -H z.B. $\text{M}-\text{CH}_2-\text{Cl}$, $\text{M}-\text{CH}_2-\text{OCH}_3$, $\text{M}-\text{CH}_2-\text{Ph}$
- sterische Maßnahmen
 - β -H von M fernhalten $\text{M}-\text{C}\equiv\text{CH}$, $\text{M}-\text{CH}=\text{CH}_2$
 - Ringspannung im Produktolefin $\text{M}-\text{C}_6\text{H}_5 \rightarrow \text{M}-\text{H} + \text{C}_6\text{H}_6$

- koord. gesättigte M ohne freie Koord.stelle (durch stark gebundene L)



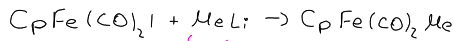
- Voraussetzung:
- freie Koord.stelle an M, cis zu Alkyl
 - planare MCCH -Anordnung möglich
 - für ÜM idR schneller als für HG-Metalle
 - vermutl. konzentrierter Verlauf

b) Reduzierende Eliminierung

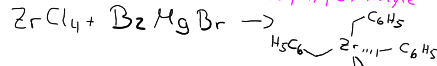


Synthesen

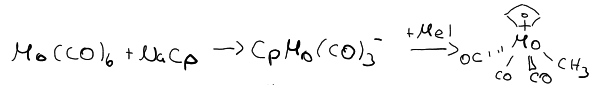
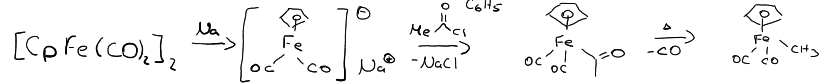
a) $MX + R^-$: nuc. Angriff an M



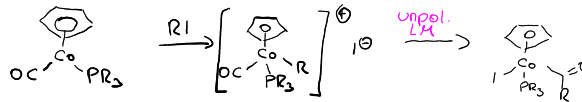
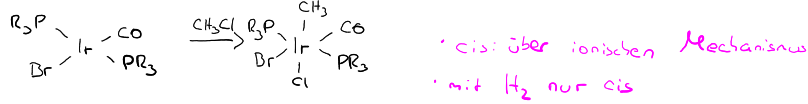
(Li- & Mg-Alkyle reaktiv, aber pot. Red. Mittel)
Alternative: Al-, Sn-, Zn-Alkyle $\rightarrow$ kovalenter



b) $M^+ + R^-$: el. Angriff an M



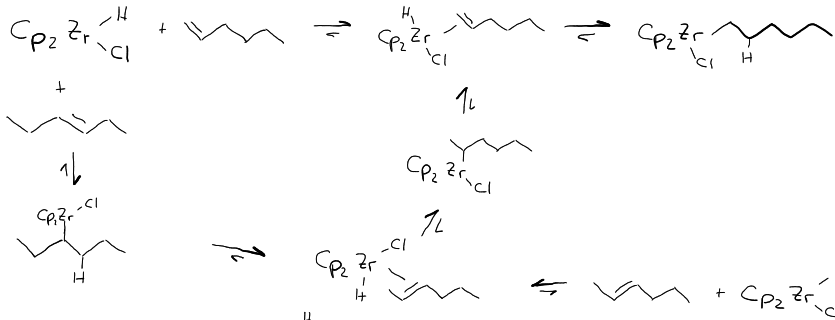
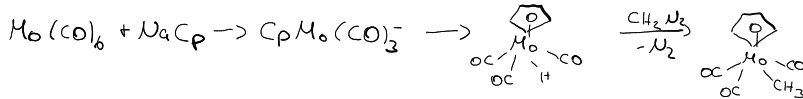
c) ox. Addition



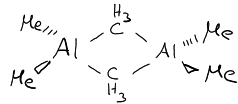
d) nuc. Addition an Olefinkomplexe



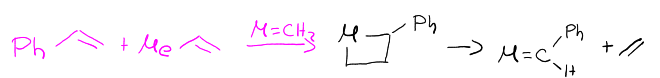
e) Insertion



• Alkylgruppen brückenständig

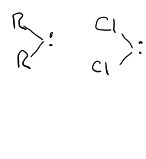
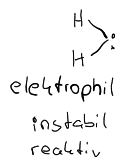


• Metallacyclobutanen



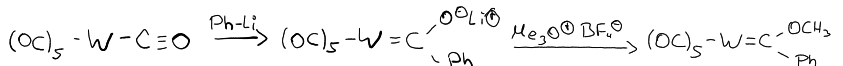
Komplexe mit M-C-Mehrfachbindungen

$M=C<$ Carben
 $M=C=C<$ Vinyliden
 $M \equiv C-$ Carbin
Alkyliden

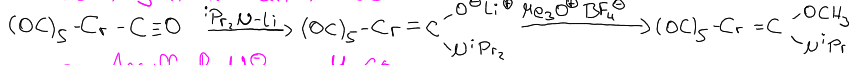


Fischer-Carbene

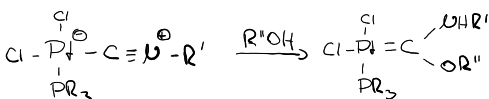
- Donor-Atom am Carben-C
- niedrige OZ am Metall
- Akzeptorliganden (CO) am Metall
- Carben-C elektrophil, δ^+



nuc. Angriff R^+ an M-CO



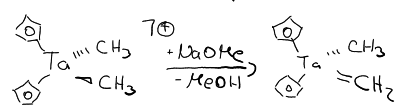
nuc. Angriff R_2N^+ an M-CO



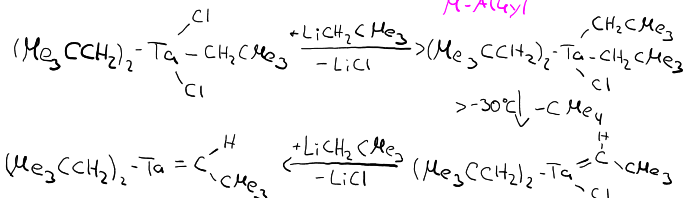
nuc. Angriff ROH an Isocyanid-Komplex

Schrock-Carbene

- C oder H an Carben-C
- hohe OZ am M
- Donorliganden am M (Cp, Hal, PR_3)
- Carben-C nucleophil

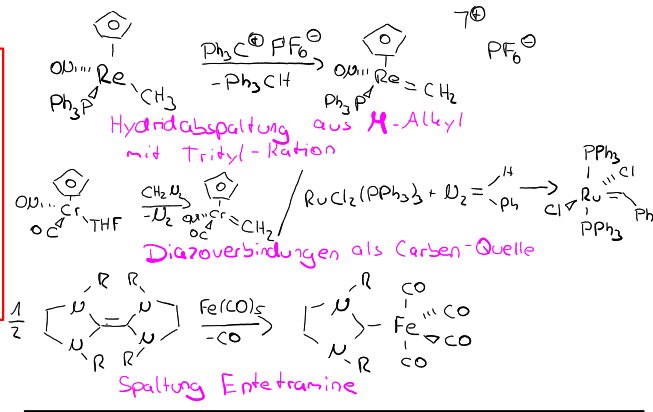


Deprotonierung von M-Alkyl

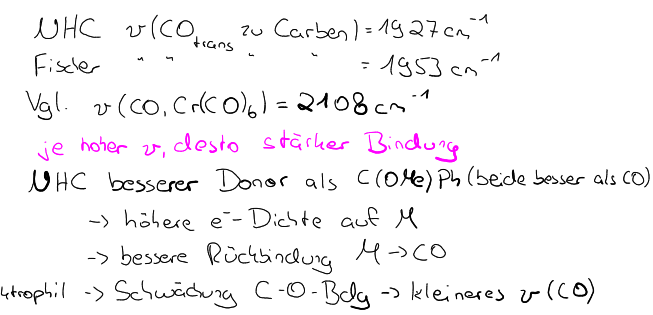
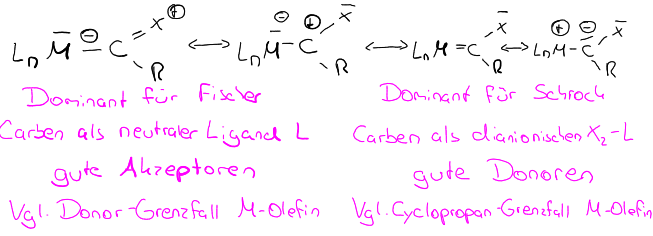
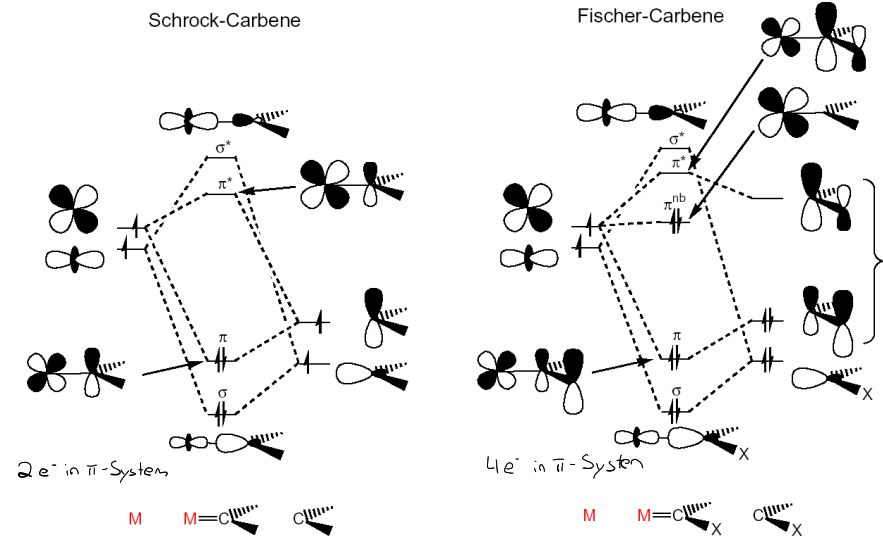


Klassifizierung

	UHC-Komplex	Fischer-Carbenkomplex	Schrock-Carbenkomplex
OZ M	beliebig	niedrig	hoch
Co-Liganden an M	beliebig	π -Akzeptoren (CO)	Donoren (Cp, PR ₃ , Hal)
Reste an Carben-C	1/2 U-Atome	min 1 Heteroatom	Alkyl, H
Reaktivität an Carben-C	inert	elektrophil	nucleophil
M-C-DB-Charakter	schwach	mittel	hoch
M-Carben-Rückbindung	niedrig	mittel	hoch



Bindungsverhältnisse & Rückbindungsanteil



HOMO an Carben-C → Carben: δ^- , nucleophil

LUMO an M

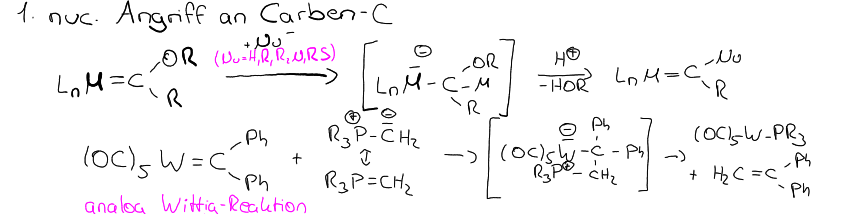
HOMO an M

LUMO an Carben-C → Carben: δ^+ , elektrophil

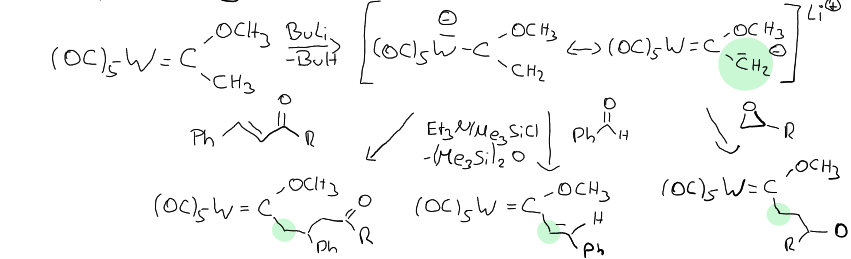
→ Schwächung C-O-Bdg → kleineres $\nu(\text{CO})$

Reaktivität

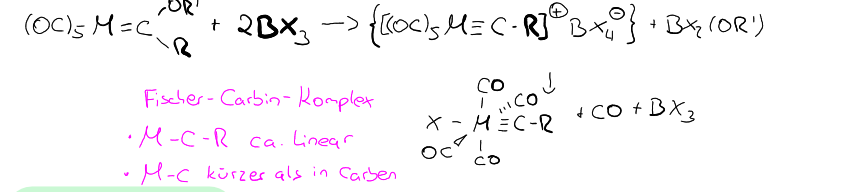
Fischer-Carbenkomplexe



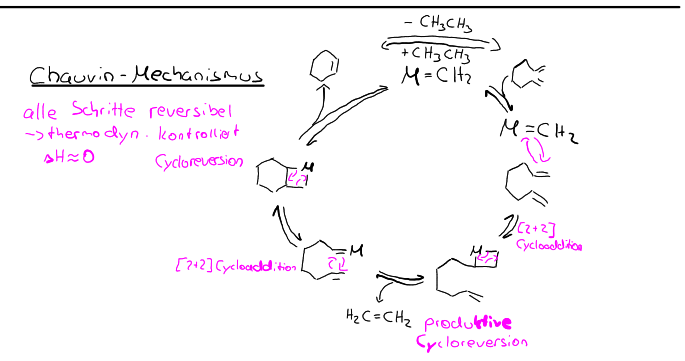
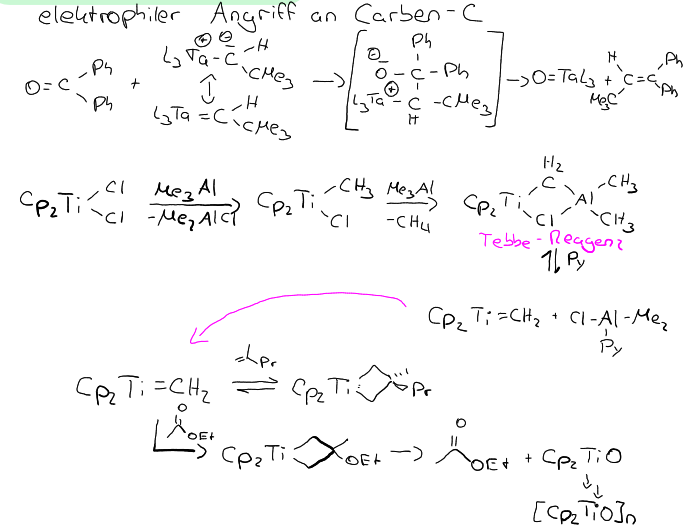
2. Deprotonierung + Elektrophil



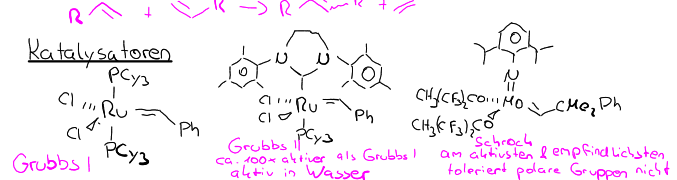
3. Abstraktion von RO⁻ durch Lewis-Säuren



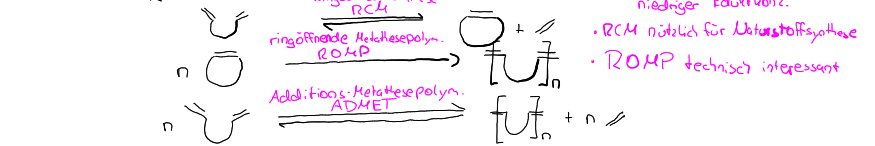
Schrock-Carbenkomplexe



Olefinmetathese



Varianten:

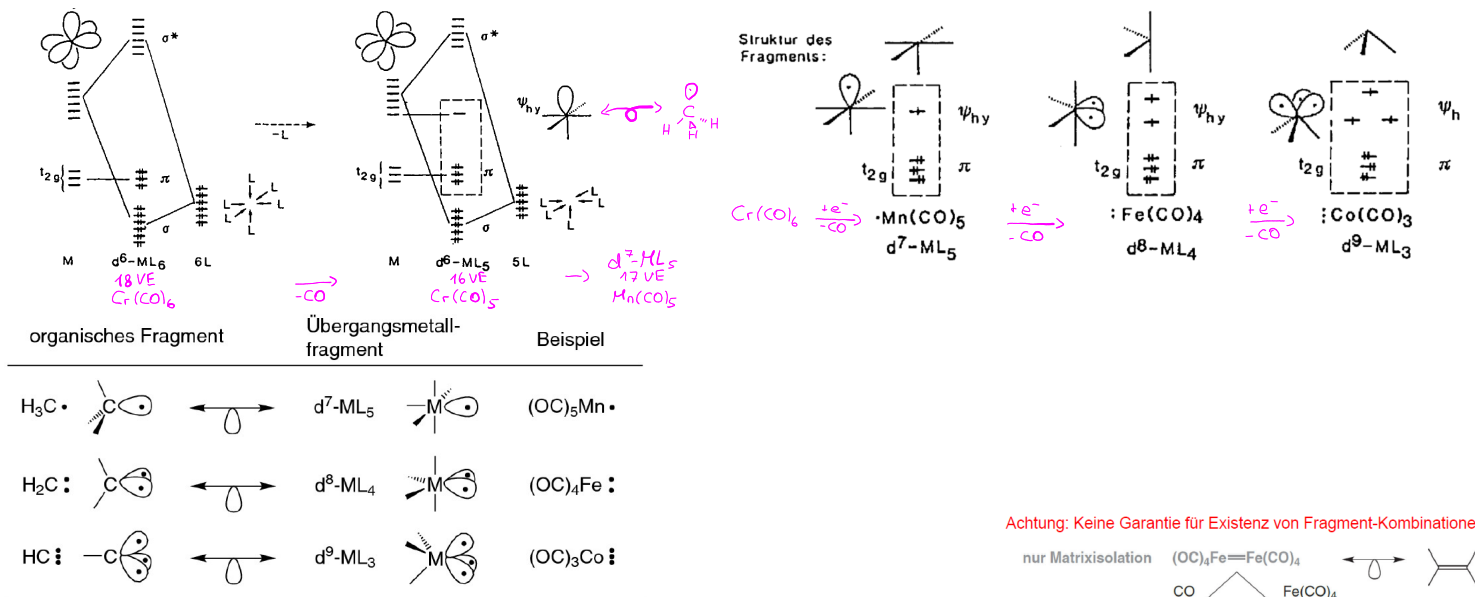


$$\text{H}_2\text{C}=\text{C}(=\text{O}) + 2 \text{R-NH}_2 + \text{H}_2\text{CO} \longrightarrow \text{Imidazolium-Kation} \xrightarrow{-\text{H}^+} \text{Imidazol-2-yliden}$$

- 1) Stöchiometrische Zusammensetzung von Verbindungen - Oktettregel, PSE, 18 VE-Regel (hohe Ligandenfeldaufspaltung)
- 2) Struktur - VSEPR-Regel
- 3) Reaktivität - HSAB, Brønstedt, Arrhenius, Lewis, Elektronegativität; MO-Theorie; Struktur-Wirkungsbeziehungen

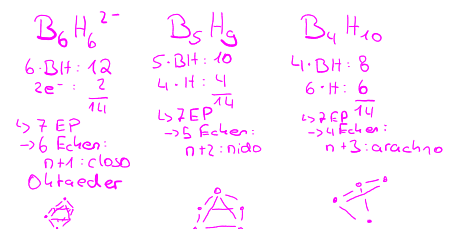
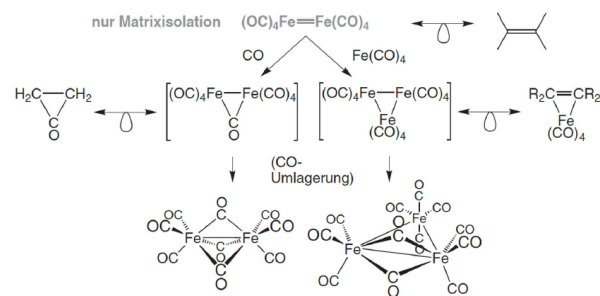
- Anzahl
- Symmetrieeigenschaften
- Energie
- räumliche Gestalt
- Grenzorbitale
- Zahl e^-

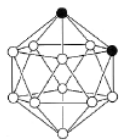
ähnlich sind



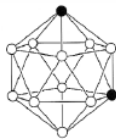
$H_3C: \xrightarrow{\sigma} d^7-ML_5 \rightarrow (OC)_5 Mn-Mn(CO)_5 \xrightarrow{\sigma} (OC)_5 Mn-CH_3 \xrightarrow{\sigma} H_3C-CH_3$
 $H_2C: \xrightarrow{\sigma} d^8-ML_4 \rightarrow (OC)_4 Os \rightarrow Os_2(CO)_4 \xrightarrow{\sigma} H_2C=C(CH_3)_2$
 $HC: \xrightarrow{\sigma} d^9-ML_3 \rightarrow R_2C=C(R)R \rightarrow R_2C=C(R)R \rightarrow (CO)_3Cr \rightarrow R_2C=C(R)R$
 $H-B: \xrightarrow{\sigma} :C \equiv H \xrightarrow{\sigma} :C \equiv N \xrightarrow{\sigma} H-C \equiv N \xrightarrow{\sigma} H-C \equiv P$
 Phosphaalkin

Cluster mit n -Ecken: $n+1$ Gerüst- e^- -Paare $\rightarrow$ cluso | Abnahme e^- -Mangel
 $n+2$ " " " $\rightarrow$ nido | Zunahme Öffnung
 $n+3$ " " " $\rightarrow$ arachno

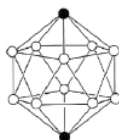




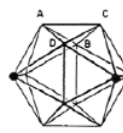
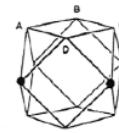
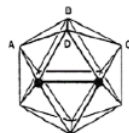
ortho



meta

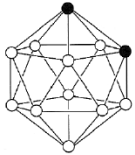
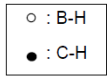


para



keine Direktsynthese möglich
thermische Umlagerung
Diamond - Square - Diamond Umlagerung

Clusterabbau von Carboranen



1,2-C₂B₁₀H₁₂

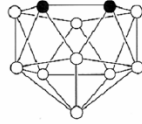
1. NaOEt

- B(OEt)₃

- H₂

2. NaH

- H₂



C₂B₉H₁₁²⁻

„Cooking pot“

B-H: 2 Elektronen

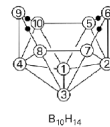
C-H = B-H: 3 Elektronen

isolobal zu Cp⁻
13 EP/11 Ecken
→ nido

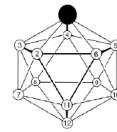
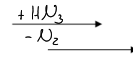
13 EP/12 Ecken
→ closo

Azaborane

N-H als Clusterecke, Nitren: $\text{N-H} \leftarrow \text{O} \rightleftharpoons \text{C}^{\ominus}\text{-H} \leftarrow \text{O} \rightleftharpoons \text{B}^{\oplus}\text{-H}$



B₁₀H₁₄



NB₁₁H₁₂

13 EP/12 Ecken
→ closo

